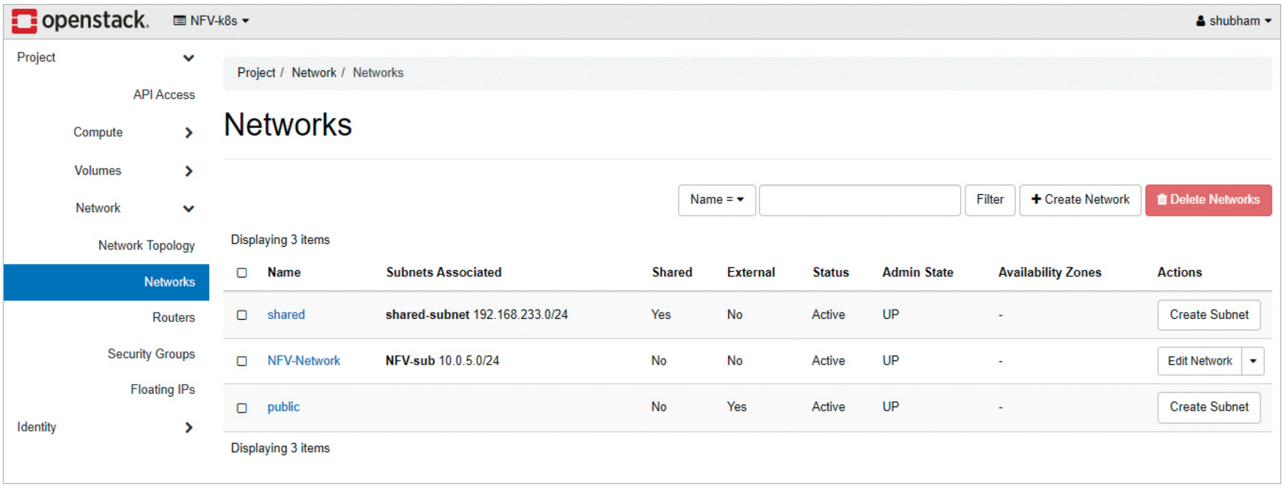

Figure 2: Networks over OpenStack for the NFV-enabled Kubernetes cluster

configurations throughout the environment while ensuring that similar packages and dependencies are installed. In this pipeline, we have to install the NFV containers over our infrastructure to make them readily available throughout the network. Ansible deploys the application and manages the related configuration via playbooks written in YAML. The playbooks communicate through SSH to all the hosts written in the inventory file.

A similar playbook is written for the deployment, where various Kubernetes configuration files are registered by the Ansible server to the deployment machines. Various Kubernetes objects are applied via the declarative methodology using the configuration files at *https://github.com/shubhamaggarwal890/nginx-vod/blob/master/deploy-kubernetes-config.yml*.

Deployment over Kubernetes and OpenStack

Infrastructure is the key aspect of any deployment, especially when we talk about NFV-enabled infrastructure. We have seen how Kubernetes helps in the enablement of NFV and provides resources for it to be scalable and self-healing. We wanted an environment that enables horizontal as well as vertical scalability. Horizontal scalability refers to the addition of new nodes or virtual machines to the current infrastructure setup, whereas vertical scalability refers to the addition of computation, networking, and storage resources to an existing node or virtual machine. With Kubernetes, this is successfully achieved where the number of replica deployments can be increased imperatively via Kubernetes internal tools, and further, the cluster size of Kubernetes can be increased too by adding new virtual machines through OpenStack.

After the installation of OpenStack, we tailored security groups and networking specifically to the needs of Kubernetes and deployed multiple VMs for it to entertain the Kubernetes cluster. These nodes connect and achieve orchestration, over which our NFV application is deployed. To enable the deployment of NFV containers and create services, the Kubernetes configurations files are written, which can be found in the NFV source code repository at *https://github.com/shubhamaggarwal890/nginx-vod/tree/master/deploy-kubernetes*.

Figure 2 illustrates the creation of a custom network and its subnet specifically for our NFV module over the OpenStack platform, where all the network services are abstracted through the Neutron component. Various details such as DNS, Class address, and IP range are decided while creating the network. The Kubernetes cluster is then connected to this custom network allowing communication between its various nodes.

After the successful establishment of the network between the nodes of the Kubernetes cluster, we need to provide access to the internet to the cluster nodes. This connection is required so that the Kubernetes cluster can patch new updates, and download the Docker images from the Docker registries. Our custom network is connected with the public/external network via a router. This router is created by OpenStack and linked to the interface of our custom network, enabling the flow of traffic to the internet.

OpenStack enables security groups for every running instance. In the security group, administrators can manage the security rules and the flow of traffic. Figure 3 showcases the enabled security group for the project, where all the Kubernetes-related traffic is allowed to flow through different ports and protocols.

Various instances, as shown in Figure 4, are created to facilitate the Kubernetes cluster. These instances are attached to the respective networks, security key pairs, and security groups. A floating IP is attached to these instances, which provides a